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import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.tree import DecisionTreeClassifier, export_text

df = pd.read_csv("iris.csv")

X = df[['sepal_length', 'sepal_width', 'petal_length', 'petal_width']]
Y = df['species']

X_train, X_test, Y_train, Y_test = train_test_split(X, Y,
test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

k = 5
knn = KNeighborsClassifier(n_neighbors=k)
knn.fit(X_train, Y_train)

KNeighborsClassifier()

Y_pred = knn.predict(X_test)

print("Accuracy:", accuracy_score(Y_test, Y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(Y_test, Y_pred))

```

Accuracy: 1.0

Confusion Matrix:

```

[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

```

```

df = pd.read_csv("iris.csv")
df

```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
..	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

```

[150 rows x 5 columns]

dtree = DecisionTreeClassifier(criterion='gini', max_depth=3,
random_state=42)
dtree.fit(X_train, Y_train)

DecisionTreeClassifier(max_depth=3, random_state=42)

Y_pred = dtree.predict(X_test)

print("Accuracy:", accuracy_score(Y_test, Y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(Y_test, Y_pred))

import matplotlib.pyplot as plt
from sklearn.tree import plot_tree

plt.figure(figsize=(15, 10))

plot_tree(
    dtree,
    feature_names=X.columns,
    class_names=dtree.classes_,
    filled=True,
    rounded=True
)

plt.show()

Accuracy: 1.0

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

```

